

Contact

New York City, NY, USA.
+1 412-778-9787 (Mobile)
sahilkumar4ai@gmail.com

www.linkedin.com/in/sahilkumar4ai
(LinkedIn)
github.com/sahilkumar15 (Other)

Top Skills

Embedded Linux
Linux
Aagentic AI

Languages

Hindi (Native or Bilingual)
English (Full Professional)

Certifications

Data Analyst

Sahil Kumar

AI Research Scientist | PhD in AI | LLMs, Generative AI, CV | First
Author - ICLR, INTERSPEECH
New York, New York, United States

Summary

I'm a PhD candidate in Mathematics and Artificial Intelligence with over 6 years of experience designing and deploying advanced AI solutions. My expertise spans Large Language Models (LLMs), Generative AI, Deep Learning, and Computer Vision, with a strong foundation in data structures, algorithms, and applied machine learning.

Currently, my research focuses on transformer-based architectures and multi-modal AI systems. Our latest work, ViTVS, introduces a novel audio denoising approach using Vision Transformers and has been accepted at INTERSPEECH 2024. Read the paper

With hands-on experience in tools like TensorFlow, PyTorch, Hugging Face, LangChain, OpenAI, Gemini, and Vector DBs, I specialize in building end-to-end AI pipelines—from data preprocessing and model training to cloud deployment using Azure, AWS, and GCP.

I enjoy solving complex problems at the intersection of theory and real-world application. Whether it's developing research code, optimizing machine learning workflows, or publishing impactful research, I'm passionate about pushing the boundaries of what AI can do.

Experience

Yeshiva University
Research Assistant
January 2024 - Present (2 years 6 months)
New York, New York, United States

•Conducting cutting-edge research in Legal Machine Learning, Generative AI, Computer Vision, and NLP to improve decision-making and automation in legal tech.

Developing Digital Human Modeling (DHM) systems for immersive, interactive educational platforms, integrating state-of-the-art models with human-computer interaction.

Publishing and presenting at top AI conferences including INTERSPEECH 2024.

Tools & Technologies: Python, PyTorch, LangChain, OpenAI, Gemini, Vision Transformers

Tata Consultancy Services

Senior Data Scientist

January 2023 - December 2023 (1 year)

Designed predictive models for ad optimization, boosting Apple's ad revenue by 25% and targeting efficiency by 30%.

Utilized BERT, Transformers, and LDA sentiment analysis under data storage constraints, ensuring scalable ML deployment.

Recognized for problem-solving, team collaboration, and clear technical communication.

Tools & Techniques: BERT, Transformers, LDA, Python, ML Pipeline Optimization

Tiger Analytics

Senior Analyst Data Scientist

September 2021 - December 2022 (1 year 4 months)

India

Improved team productivity by 30% through workflow optimization and promoted statistical best practices across EDA, CV, and DL pipelines.

Mentored junior scientists, resulting in a 50% improvement in advanced modeling proficiency.

Led cross-functional projects involving experimental design, CV models, and statistical modeling.

Tata Elxsi

3 years

Senior Engineer Data Scientist

September 2020 - July 2021 (11 months)

Bengaluru, Karnataka, India

Key Achievements

Enhanced Model Performance: Achieved a 75% improvement in model accuracy and efficiency by applying advanced machine learning algorithms and statistical methodologies, resulting in significantly more reliable and robust predictive outputs.

Custom Data Pipeline Development: Designed and implemented end-to-end, modular data pipelines, leading to a 25% increase in system performance by optimizing data ingestion, transformation, and preprocessing stages.

Impactful Communication & Visualization: Effectively conveyed complex analytical insights to non-technical stakeholders, resulting in a 40% improvement in understanding. Created intuitive reports and dashboards that supported informed decision-making at all levels.

Core Responsibilities

Developed, tuned, and deployed machine learning models tailored to business-critical challenges using a combination of statistical and deep learning techniques.

Built scalable and reusable data pipeline architectures to ensure seamless data flow for training, validation, and production environments.

Translated analytical findings into actionable insights using visual storytelling and presented them to cross-functional teams and senior leadership to influence strategic direction.

Skills Used

Python, Scikit-learn, Pandas, SQL, Matplotlib, Seaborn, Tableau, Machine Learning, Statistical Modeling, Data Pipeline Architecture, Data Communication, Business Storytelling

Data Scientist

August 2018 - September 2020 (2 years 2 months)

Bengaluru Area, India

Key Responsibilities & Contributions

Cross-Functional Collaboration: Partnered with stakeholders across departments to identify strategic opportunities for leveraging data, leading to the development of impactful, business-aligned analytics solutions.

Data Quality Evaluation: Assessed the reliability and accuracy of newly acquired data sources and collection methods, ensuring high-quality inputs for analysis and model development.

Custom Model Development: Designed and implemented tailored data models and algorithms for diverse datasets, significantly improving the depth, precision, and business relevance of analytical insights.

Predictive Analytics Application: Deployed predictive models to optimize customer experience, enhance ad targeting effectiveness, and support revenue generation through data-driven marketing strategies.

Performance Monitoring & Data Integrity: Established robust workflows, advanced techniques, and monitoring tools to evaluate model performance while maintaining strict standards for data consistency and accuracy.

Aeronautical Development Agency

Machine Learning Engineer

January 2018 - June 2018 (6 months)

Bengaluru Area, India

Key Responsibilities & Contributions

NLP Application Design: Conceptualized and developed Natural Language Processing (NLP) applications, with a strong focus on transforming unstructured text into meaningful, actionable features through effective text representation strategies.

Algorithm Selection & System Development: Identified and implemented the most suitable algorithms and tools for diverse NLP tasks. Developed complete NLP systems tailored to specific project goals, applying the most relevant frameworks and methodologies for optimal performance.

Model Training & Evaluation: Led end-to-end model training workflows and conducted thorough evaluation experiments to assess performance. Customized and extended machine learning and deep learning frameworks to ensure high accuracy and efficiency in NLP model deployment.

Education

Yeshiva University

Doctor of Philosophy - PhD, Mathematics and Computer Science · (August 2024 - May 2027)

Yeshiva University

Master's degree, Artificial Intelligence · (January 2023 - August 2024)

Indian Institute of Technology, Roorkee

Postgraduate Degree, AI for Digital Businesses and Ecommerce · (2020 - 2021)

Indian Institute of Technology, Madras

Bachelor's degree, Data Science and Applications · (2014 - 2018)

Delhi Public School

Intermediate, PCM · (2013 - 2014)